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Data-Driven Method Enhancing Wellbore Cleanup Efficiency

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Abstract

Focused on cleaning and priming wellbores, this paper shows how a recently developed cleanup system fosters more efficient operations through a data driven method. The method has been applied to more than 450 wells in China, dramatically raising operation efficiency more than 51%.

The method addresses operators' needs concerning cleanup tool assembly, working fluid, and design protocols. Diagnostic solutions that carrying data to the surface through recorded sensors, by which operators can obtain a nuanced comprehension of wellbore conditions. A suggestion would be recommended, which providing a friendly decision environment. Guided by data-driven decisions, tools and working fluids can be adjusted in subsequent processes. To meet different decision requirements, scrapers and other downhole tools with sensors are optimized, and two types of working fluids are presented.

The approach, referred to as flex-cleanup, prioritizes simplicity while offering operators the necessary components to engineer a wellbore faced to specific requirements. It is designed to be adaptable, catering to both minimalistic and intricate needs for completely cleaning.

The method's adaptability renders it suitable for operators' dealing with various types of wellbores, be they vertical or horizontal. Nevertheless, certain factors and conditions further augment the appeal of this solution. For instance, it becomes particularly attractive when faced with challenges posed by traditional complex casing strings, or in scenarios involving both cement milling and casing scraping. It is found that the novel method can dramatically improve operation efficiency.

This method to cleanup wellbores presents the industry with an alternative, leading to a reduction in unnecessary procedures, streamlined operations, and decreased equipment deployment, ultimately resulting in shorter operation times and safer working conditions.

Introduction

Against the backdrop of continuously rising global energy demand, the oil and gas industry is pursuing increasingly operational efficiency and safety standards, a trend that has strongly driven continuous innovation in wellbore construction and intervention technologies. Among them, wellbore cleanup (WBCU)

as a fundamental process occupies an irreplaceable position in the entire oil and gas exploration and development process. Its core task is to remove debris such as drilling fluids, cuttings, and cement residues accumulated during the drilling and completion operations. If these substances remain in the wellbore for a long time, they will trigger a series of chain downhole problems.

Specifically, residual impurities will increase the friction when subsequent tools are run into the wellbore, which may not only damage the tools but also prolong the operation time, significantly increasing operational risks and costs. Statistics show that between 2021 and 2024, the incidence of stuck pipe and plugging accidents in wellbore cleanup operations in southwest China's oil fields reached as high as 15%, a figure that fully highlights the urgent need for efficient wellbore cleanup technologies. In addition, an insufficiently cleaned wellbore may cause formation damage, because trapped solid particles will be pushed into formations after perforation and block formation pores, which affects extraction efficiency.

Traditional wellbore cleanup methods, although able to meet basic operational requirements to a certain extent, have gradually exposed many drawbacks when facing complex and variable wellbore conditions, such as complex operational processes, low cleanup efficiency, and limited adaptability. These traditional methods usually rely on fixed tool configurations, making it difficult to flexibly adjust according to the actual conditions of different wellbores; data acquisition is nearly zero or mainly depends on cables, which not only limits the operating range of the tools but also increases the difficulty of equipment maintenance and troubleshooting; at the same time, the adoption of a "one- fits- all" fluid system makes it impossible to achieve precise cleaning for different geological environments and wellbore characteristics.

As oil and gas exploration and development continue to advance into deeper and more complex formations, wellbore structures have become increasingly complex, with horizontal wells, extended- reach wells, and multilateral wells and other new wellbore structures gradually increasing, and the limitations of traditional methods have become more obvious. Traditional cleaning methods suitable for vertical wells may have poor effects in horizontal section casings, which easily lead to impurity residues and seriously affect completion efficiency. To solve problems existing in traditional wellbore cleanup methods, the industry has been actively exploring simpler and more adaptable solutions. Based on this, this paper proposes an innovative wellbore cleanup method "flex - cleanup". This method aims to balance operational simplicity and cleaning performance. Unlike traditional rigid and "one- fits- all" systems, "flex - cleanup" adopts a modular tool design, which can be quickly combined and adjusted according to different wellbore conditions to improve tool adaptability; with the help of diagnostic technology, it enhances the flexibility and safety of operations; at the same time, through customized fluid engineering design, it can optimize the performance of cleaning fluids for specific wellbore environments to achieve efficient and precise cleaning. Whether for simple operation scenarios such as shallow vertical wells with simple structures or complex scenarios such as multilateral horizontal wells with damaged casings, "flex - cleanup" can effectively meet their wellbore cleanup needs, providing new ideas and methods for improving the efficiency and quality of wellbore cleanup.

Data-driven Method

Traditional Wellbore Cleanup (WBCU) techniques rely heavily on qualitative evaluation. In specific operations, scrapers are run up and down 3 to 5 times at designated depths, but there is a lack of quantitative criteria for assessing wellbore conditions throughout the process. This qualitative evaluation method often can only judge the cleaning effect based on the experience of operators, making it difficult to accurately grasp the actual situation inside the wellbore. It is prone to problems such as incomplete cleaning or over - cleaning, which in turn affects the smooth progress of subsequent operations.

However, data plays a crucial role in wellbore cleanup operations. It can monitor the inner diameter of the casing in real - time, promptly detecting potential problems such as deformation and wear of the casing. It evaluates the service life of operating equipment, providing a scientific basis for equipment maintenance

and replacement. Meanwhile, it can also assess working fluid conditions by measuring parameters such as temperature and pressure. Through comprehensive analysis of these data, the cleaning efficiency can be accurately evaluated, providing strong support for optimizing the cleaning process and improving the cleaning effect. In practical applications, the parameters mostly used are as follows:

- **Casing ID** → Detects wellbore deformation.
- **Hook load variation** → Predicts debris accumulation and potential sticking points.
- **Temperature & pressure** → Evaluates working fluid status.
- **Acceleration** → Evaluates tool vibration.

The Data-Driven approach leverages these parameters to optimize subsequent operations for the target well and analogous wells in the same field. This is particularly vital in extended-reach (ERD) and horizontal wells, allowing determination of casing geometry integrity, necessity of cleanup trips or sand washing, etc. Among all the parameters, the most complex one is wellbore diameter measurement, which employs ultrasonic displacement sensors to track scraping blade movement. Algorithm specifics:

Given four points on a circle of radius r , each moves toward the center by distances a, b, c, d . Find the diameter of the best-fit circle for the four new points, as Fig.1(a) and (b) shows.

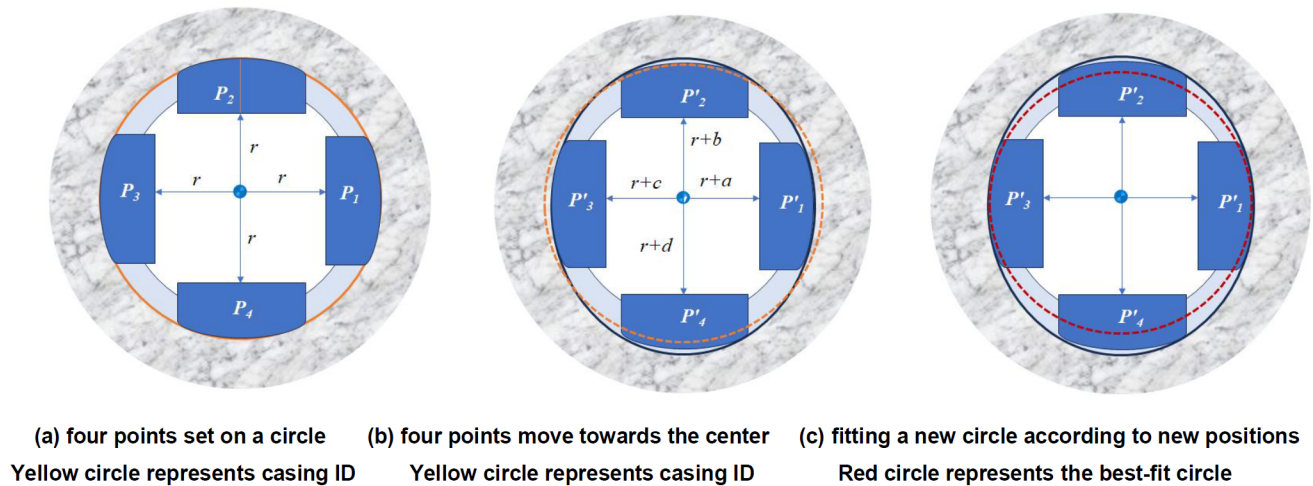


Figure 1—Scheme of measuring casing ID by fitting four points

Place the circle at the origin. The four initial points are $(r\cos\theta_i, r\sin\theta_i)$. After moving inward/outward radially, the new points are:

$$P'_i = ((r + d_i)\cos\theta_i, (r + d_i)\sin\theta_i)$$

Here, we choose four points at $0^\circ, 90^\circ, 180^\circ, 270^\circ$.

$$P'_1 = (r + a, 0)$$

$$P'_2 = (0, r + b)$$

$$P'_3 = (-(r + c), 0)$$

$$P'_4 = (0, -(r + d))$$

As Fig.1(b) shows, the circle equation $(x - h)^2 + (y - k)^2 = r^2$ yields Center Position.

$$h = \frac{c - a}{2}, k = \frac{d - b}{2}$$

and radius is

$$r = \frac{1}{2} \sqrt{(2r - a - c)^2 + (b - d)^2}$$

It should be noted that this method cannot accurately measure the inner diameter of the casing, but it can reflect abrupt changes in the inner diameter. Special attention should be paid to abnormal locations in the data. Running resistance is determined by hook load. Temperature and pressure are measured by corresponding microsensors.

Flex-Cleanup: Design Principles and Technical Components

Flex-cleanup is built around three core principles: simplicity, adaptability, and cost-effectiveness. The most important are technical architecture, including modular tooling, cable-free diagnostics, and fluid engineering.

Modular Tool Architecture: Versatility Through Standardization

At the heart of flex-cleanup is a modular tool string composed of interchangeable components, enabling rapid reconfiguration for specific cleanup tasks. The system's core modules include:

Scrapers with microsensors: Scrapers feature spring-loaded blades with a 10 to 15% diameter range, accommodating minor casing deformation (up to 5% of nominal diameter) and irregularities. Blades are fabricated from high-strength alloy steel to withstand abrasion from cement and scale. Spring force is specially adjusted according to estimated well condition to ensure a thoroughly cleaning. Fig.2 shows casing inner wall condition before and after scraping.



Figure 2—Casing inner wall condition before and after scraping

Four microsensors are installed directly beneath the blades, utilizing microwave technology to measure displacements. Time and displacement data are stored in downhole recorders, and the inner diameter of the casing can be fitted by analyzing the data in accordance with the method specified in Section 2.

Magnet with temperature sensor: Temperature significantly affects the performance of magnets, as common magnets typically have an operating temperature below 180°C. To grasp the actual downhole working conditions, a miniature temperature sensor is mounted at the centralizer position of the magnet tool to measure and store downhole temperature.

Centralizers: Low-friction bow-spring centralizers ensure the centering of the tool assembly in horizontal wells, enabling even distribution of contact pressure across the casing walls. This minimizes eccentricity-induced cleaning gaps, especially in high-angle (>60°) well sections where gravity would otherwise pull the tools toward the low side.

This modularity allows operators to configure tool assembly for specific challenges: a "light" configuration (scraper + centralizer) for routine cleaning in vertical wells, or a "heavy" configuration (scraper + mill + centralizer + magnet) for post-cementing cleanup in horizontal section.

Data-Driven Decision-Making

Flex-cleanup employs a self-contained data logging system, eliminating surface cables and enabling greater operational flexibility. The diagnostic module includes sensors capturing data on casing condition, fluid properties, and tool performance. Fig.3 shows main sensors in the tool and a continuous decision process. Firstly, the overall objectives of the process are defined as cleanup efficiency and operational safety. These objectives guide the selection of methods, which in turn affect the entire workflow. The core of the process is data-driven work, which is supported by key elements, data, and processes. Key elements refer to the essential components and factors crucial to the process, as casing integrity, fluid properties, and tool performance. Data is the raw material input into the system, and processes are procedures followed. Within the data-driven work module, there is a crucial step called data filtering and decision-making. This involves filtering incoming data to ensure its quality and relevance, and then making informed decisions based on the filtered data. The output of the data-driven work module, including the results of data filtering and decision-making, is then evaluated against the "fit for purpose scenario". The scenario is always a kind of regional demand, such as the need to increase efficiency and quality in a specific oilfield block. This means checking whether the results meet the specific requirements and objectives of the intended application or situation. Finally, the results that pass the "fit for purpose scenario" evaluation are distributed, probably to relevant stakeholders or for further use.

- Casing integrity (best-fit curve about casing ID from displacement measurement, hook-load variation).
- Fluid properties (downhole temperature, pressure).
- Tool performance (acceleration, blade wear).

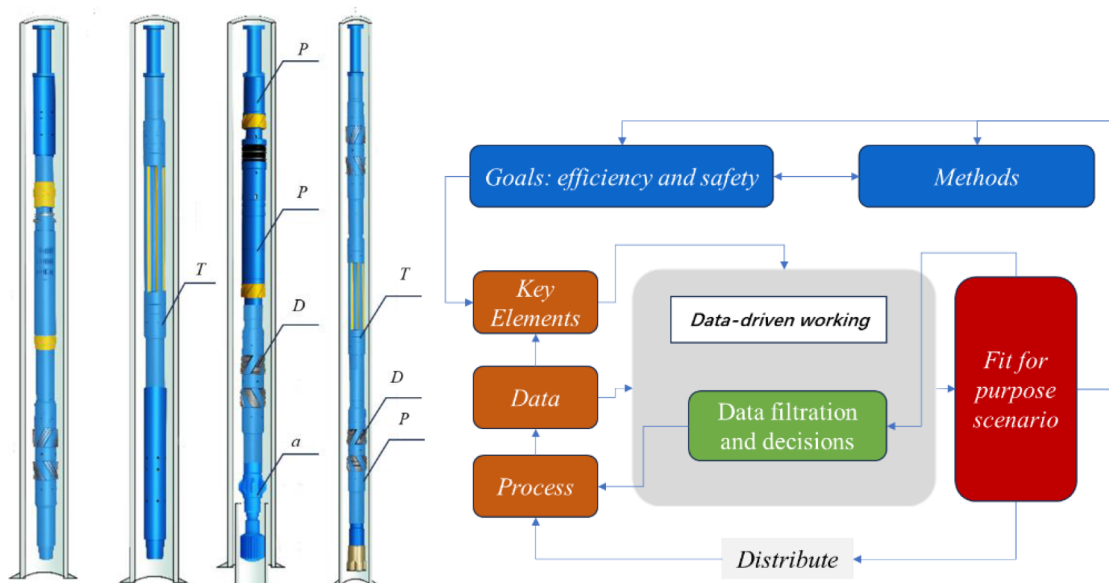


Figure 3—Flex-Cleanup tool assembly scheme T, P, D, a are temperature, pressure, displacement and acceleration sensors.

Data Storage: A ruggedized, high-capacity memory module records data at 10Hz of continuously. Storage is rated for temperatures up to 175°C and pressures up to 15kpsi, suitable for deep and high-temperature wells.

Retrieval Mechanism: Post-operation, data is retrieved via drillstring during tool recovery. Data is processed using proprietary software that generates visualizations of casing condition and cleanup efficiency, highlighting areas requiring rework.

Working Fluid Optimization

Flex-cleanup's fluid system is engineered to complement mechanical cleaning, enhancing debris transport and minimizing formation damage. The fluid formulation is customized using a three-step framework:

- **Condition Assessment:** Based on well logs and mud reports, operators identify contaminants and formation sensitivity.
- **Base Fluid Selection:** Options include water-Based Fluids (WBFs) and Oil-Based Fluids up to 180°C.
- **Hybrid Fluids:** Emulsified WBF-OBF blends for wells with mixed contaminants.



Figure 4—Working fluid and testing

Fluid circulation rates are optimized using computational fluid dynamics (CFD) models, ensuring annular velocities exceed the critical transport threshold in all well sections. In horizontal laterals, where gravity segregates solids, spiral flow generators induce swirling lifting debris into the high-velocity fluid core. This addresses the longstanding challenge of solids settling in low-side regions, even when average annular velocities meet vertical well standards.

Operational Workflow

Flex-cleanup operations follow a streamlined, four-phase workflow designed to minimize NPT:

Pre-Job Planning. Wellbore characterization via logs and offset data to identify contaminants, casing dimensions, and deviation profiles. Tool assembly configuration by simulating tool performance in the target well trajectory to identify potential sticking points. Fluid formulation testing in a lab environment.

Tool Deployment. Select a light or heavy-duty configuration, rig-up of the modular tool assembly integrated. Initial circulation at low rate to prime the system, followed by incremental rate increases to target velocity.

Cleaning and Data Acquisition. The tool assembly is run in hole (RIH) at 30–50 ft/min, with mechanical cleaning and fluid circulation occurring simultaneously. The sensors records data continuously.

Post-Job Analysis. Data retrieval and processing to generate a cleanup efficiency report. If required, a second run with adjusted tooling/fluid parameters is executed, based on data insights. Similar wells in this area could be operated in the same scheme. This workflow reduces the need for redundant steps—such as separate logging runs to verify cleanup—by integrating data acquisition directly into the cleaning process.

Field Case Studies

To validate flex-cleanup's performance, this section presents three field case studies spanning diverse well types, operational challenges, and reservoir conditions. Each case compares flex-cleanup to traditional methods, focusing on key metrics: operational time, cost, and cleanup efficiency.

Case Study 1: Vertical Wells with Complex Casing String

Well Characteristics: Vertical oil well in west China, well depth between 23000ft to 31000ft, completed with a tapered casing string, usually 7in. intermediate casing to 5in. liner or 7-5/8in. intermediate casing to 5-1/2in. liner. Post-drilling, the well required cleanup to remove cement laitance and drilling fluid residues before perforation.

Traditional Approach

Historical data from offset wells indicated that conventional cleanup required four tool runs. No.1 a 7-3/4in or 7in. scraper for wellbore cleaning and circulation. No.2 a drifting operation for assessing wellbore gauge. No.3 a 5 1/2in. or 5in. scraper for cleaning liner. No.4 a small drifting gauge is operated to assess liner condition, as well as displacement, pressure testing, etc. Total operational time averaged 220 hours.

Flex-Cleanup Implementation

A modular tool string was configured with scrapers (adjustable to 7-5/8in., 7in., 5-1/2in. and 5in.), magnets, brushes and a diagnostic module. The fluid system was a water-based brine with 1 vol% dispersant.

Results

A single tool run completed in 120 to 160hours, reducing operational time more than 36%. Data acquired: casing ID obtained from fitting calculation, downhole temperature and pressure, vibration from acceleration measurement. Confirmed 95% casing wall cleanliness, exceeding the 90% industry standard. Cost savings: about \$45,000 per well, primarily from reduced rig time and equipment.

Case Study 2: Horizontal Shale Well with a long liner

Well Characteristics: A 22,880ft horizontal well in southwest China, with a 9,800ft lateral. The well needs post-drilling cleanup. Drag and friction is a vital factor to the operation. More than 10 similar wells are in the same area, among which there was a sever stuck pipe downhole completion.

Traditional Approach

Offset wells requires 5-1/2in. scraper, followed by a wireline log to verify cleaning and downhole condition.

Flex-Cleanup Implementation

A tool assembly with a custom scraper (5-1/2in. nominal diameter, 15% expansion range) and a centralizer was deployed. The diagnostic module included a casing ID, hook load, and vibration recorders.

Results

A single run completed and downhole data acquired, by which maximum drag and friction zone are estimated. Cleanliness verification via diagnostic data eliminated the need for a separate wireline log. No NPT, compared to the historical 30% risk of stuck tools.

Conclusion

Wellbore cleanup is a critical yet often undervalued process in oil and gas operations, with direct implications for production efficiency and safety. Traditional methods, while functional, struggle to meet

the demands of modern well architectures—complex casing strings, horizontal trajectories, and challenging reservoir conditions—resulting in high costs, prolonged operations, and safety risks.

Flex-cleanup addresses these challenges through a deliberate focus on simplicity, adaptability, and data-driven optimization. Its modular tool architecture enables rapid reconfiguration for diverse well types. More than 450 field cases studies validate its performance: 30 to 40% reductions in operational time, significant cost savings, and enhanced cleanup efficiency across vertical, horizontal, and deepwater wells.

While limitations exist, particularly in extreme conditions and severe casing damage, ongoing innovations in automation, telemetry, and sustainability promise to expand its applicability. As the industry continues to pursue efficiency and safety, flex-cleanup represents a paradigm shift in wellbore preparation, offering a scalable, cost-effective solution that aligns with the demands of 21st-century oil and gas operations.

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